Assignment 3
Continuous Random Variables – Part 1

Course:	Probability & Statistics
Instructor:	Dr. Aamir Hasan & Dr. M. Sami Siddiqui
Total Marks:	50
Due Date:	April 2026

Submission Instructions

- Show all steps clearly. Unsupported answers may receive little or no credit.
- Write your solutions neatly and organize them by question number.
- Submit a single PDF file through Canvas.
- Submit a hard copy for the checking and it will be returned to you.

Question 1 (5 marks)
Let X have the probability density function

$$f_X(x) = \frac{\lambda}{2} e^{-\lambda|x|}, \qquad \lambda > 0.$$

(a) Verify that $f_X(x)$ satisfies the normalization condition. [1]

(b) Find the mean of X. [2]

(c) Find the variance of X. [2]

Question 2 (4 marks)
Consider a triangle and a point chosen within the triangle according to the uniform probability law. Let X be the distance from the point to the base of the triangle.

(a) Find the cumulative distribution function (CDF) of X. [2]

(b) Find the probability density function (PDF) of X. [2]

Question 3 (4 marks)
Calamity Jane goes to the bank to make a withdrawal and is equally likely to find 0 or 1 customers ahead of her. The service time of the customer ahead, if present, is exponentially distributed with parameter λ.

(a) Define a suitable random variable representing Jane's waiting time. [2]

(b) Derive the cumulative distribution function (CDF) of Jane's waiting time. [2]

Question 4 (5 marks)

Consider two continuous random variables Y and Z, and a random variable X that is equal to Y with probability p and equal to Z with probability $1-p$.

(a) Show that the PDF of X is

$$f_X(x) = pf_Y(x) + (1-p)f_Z(x).$$

[2]

(b) Consider the two–sided exponential distribution with PDF

$$f_X(x) = \begin{cases} p\lambda e^{\lambda x}, & x < 0, \\ (1-p)\lambda e^{-\lambda x}, & x > 0. \end{cases}$$

Find the cumulative distribution function (CDF). [3]

Question 5 (4 marks)

Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} ce^{-x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the constant c. [1]

(b) Find the cumulative distribution function $F_X(x)$. [1]

(c) Compute $P(2 < X < 5)$. [1]

(d) Find the expected value $E[X]$. [1]

Question 6 (6 marks)

A continuous random variable is said to have a **Rayleigh distribution** with parameter σ if its PDF is

$$f_X(x) = \begin{cases} \dfrac{x}{\sigma^2} e^{-x^2/2\sigma^2}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

(a) If $X \sim \text{Rayleigh}(\sigma)$, find $E[X]$. [2]

(b) Find the cumulative distribution function $F_X(x)$. [2]

(c) If $X \sim \text{Exponential}(1)$ and $Y = \sqrt{2\sigma^2 X}$, show that $Y \sim \text{Rayleigh}(\sigma)$. [2]

Question 7 (8 marks)

Let X have the CDF

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \le x < \frac{1}{4}, \\ x + \frac{1}{2}, & \frac{1}{4} \le x < \frac{1}{2}, \\ 1, & x \ge \frac{1}{2}. \end{cases}$$

(a) Plot $F_X(x)$ and explain why X is a mixed random variable. [2]

(b) Find $P(X \le \frac{1}{3})$. [1]

(c) Find $P(X \ge \frac{1}{4})$. [1]

(d) Write $F_X(x) = C(x) + D(x)$. [2]

(e) Find $c(x) = \dfrac{d}{dx}C(x)$. [1]

(f) Find $E[X]$. [1]

Question 8 (4 marks)

The heights of students in a Habib University classroom are approximately normally distributed. Let the deviation of a student's height from the class average be modeled by

$$f(x) = k\, e^{-x^2/2}, \quad -\infty < x < \infty,$$

where x is measured in standard units.

(a) Determine the constant k that makes $f(x)$ a valid probability density function. [2]

(b) Show that

$$\int_{-\infty}^{\infty} e^{-x^2/2}\, dx = \sqrt{2\pi}.$$

[2]

Question 9 (4 marks)

Let $X \sim N(0,1)$ and $Y \sim N(1,4)$.

(a) Find $P(X \le 1.5)$ and $P(X \le -1)$. [2]

(b) Find the PDF of $\frac{(Y-1)}{2}$. [1]

(c) Find $P(-1 \le Y \le 1)$. [1]

Question 10 (3 marks)
Let X have PDF

$$f_X(x) = \begin{cases} \dfrac{x^2+2}{3}, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find $E(X^n)$. [2]

(b) Find $\mathrm{Var}(X)$. [1]

Question 11 (3 marks)
Let $X \sim \mathrm{Uniform}(0,1)$ and $Y = e^{-X}$.

(a) Find the CDF of Y. [1]

(b) Find the PDF of Y. [1]

(c) Find $E(Y)$. [1]